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0.1 Classes-as-Types semantics

In this section, we will see, that there is a semantic of Π-, Σ-,W-, $M-$types in classes.
First of all, we (loosely) define the semantics.
Since, everything is a class in MK, we have:

$$\text{semantics}(\text{"}(A : \text{Type})\text{"}) := \top.$$

Dependent types are treated as families.

$$\text{semantics}(\text{"}(x : A \vdash B(x) : \text{Type})\text{"}) := (B \text{ family}) \wedge (\text{dom}(B) \subseteq A).$$

$$(\Lambda x : t) = \{(x, s) | (s \subseteq t) \wedge (s \text{ ss})\}$$

Is the following okey ???

$$(B \text{ family}) \longleftrightarrow \exists t : B = (\Lambda x : t)$$

0.1.1 Σ-types

In type theory, the formatin rule for a Σ-type is

$$\frac{(A : \text{Type}) \qquad (x : A \vdash B(x) : \text{Type})}{\vdash \Sigma_{(x:A)} B(x) : \text{Type}} \ \Pi\text{-form}$$

Assume that (F family).

Definition 0.1. (Dependent sum)

$$\Sigma F = \mathrm{E}^{-1} \circ F$$

0.1.2 Π-types

In type theory, the formatin rule for a Π-type is

$$\frac{(A : \text{Type}) \qquad (x : A \vdash B(x) : \text{Type})}{\vdash \Pi_{(x:A)} B(x) : \text{Type}} \ \Pi\text{-form}$$

In MK, it corresponds to the following:

Definition 0.2. (Dependent product)

$$\Pi F = \{f \in \mathcal{P}\Sigma F | (f \text{ functional}) \wedge (\mathrm{dom}(f) = \mathrm{dom}(F))\}$$

(Note: For sets the following definition is also okey:

$$\Pi F = \left\{ f \in \left(\bigcup \mathrm{ran}(F)\right)^{\mathrm{dom}(f)} \middle| \forall (x \in \mathrm{dom}(f)) : (f(x) \in F(x)) \right\},$$

but it is not working well for classes)
(Maybe TODO: improve definition so it includes families [only]. It may not be useful.)

Theorem 0.1.

0.1.3 W-types

Assume F is a family of classes. (aka "A dependent class F, which depends on $\mathrm{dom}(F)$.")

$$\mathrm{W}_{\mathrm{fun}} F(0) := \emptyset$$

$$\mathrm{W}_{\mathrm{fun}} F(n+1) := \{(r,s) | (r \in \mathrm{dom}(F)) \wedge (s : F(r) \to \mathrm{W}_{\mathrm{fun}} F(n))\}$$

$$\mathrm{W} F := \bigcup_{n \in \mathbf{N}} \mathrm{W}_{\mathrm{fun}} F(n)$$

0.1.4 M-types

$$MF := \{Q \in \mathcal{P}\{f : D \to \Sigma F | (D \in \omega) \vee (D = \omega)\} \mid$$

$\forall f, g \in Q : \forall$i initial segment of $\omega : ((\mathrm{pr}_1 \circ f) \restriction i) = ((\mathrm{pr}_1 \circ g) \restriction i) \longrightarrow f \restriction i = g \restriction i\}$